Model	Streaming	HarmB	SRLHF	BeaverTails	XSTESTR	Aegis2	WildG	Avg
LlamaGuard3-8B	no	84.5	45.2	67.9	89.8	66.1	69.5	70.5
LlamaGuard4-12B	no	83.3	42.5	68.6	88.9	63.7	66.4	68.9
WildGuard-7B	no	86.3	64.2	84.4	94.7	83.2	75.4	81.4
ShieldGemma-9B	no	60.4	44.2	62.4	86.3	70.8	49.9	62.3
ShieldGemma-27B	no	62.9	52.6	67.6	83.0	74.9	52.4	65.6
NemoGuard-8B	no	81.4	57.6	78.5	86.2	87.6	77.5	78.1
PolyGuard-Qwen-7B	no	71.1	63.3	79.5	63.4	81.9	77.9	72.9
Qwen3Guard-0.6B-Gen-strict	no	85.0	66.6	86.1	89.7	84.2	76.3	81.3
Qwen3Guard-0.6B-Gen-loose	no	82.6	64.2	85.4	91.3	84.1	77.3	80.8
Qwen3Guard-4B-Gen-strict	no	86.7	69.8	86.6	92.7	86.1	79.5	83.6
Qwen3Guard-4B-Gen-loose	no	86.7	64.5	85.2	92.4	86.5	77.3	82.1
Qwen3Guard-8B-Gen-strict	no	87.2	70.5	86.6	92.1	86.1	78.9	83.6
Qwen3Guard-8B-Gen-loose	no	86.5	64.2	85.5	93.7	86.4	77.3	82.3
Qwen3Guard-Stream-0.6B-strict	yes	83.1	62.8	84.5	84.8	81.4	76.3	78.8
Qwen3Guard-Stream-0.6B-loose	yes	80.6	61.7	84.0	83.3	81.4	75.8	77.8
Qwen3Guard-Stream-4B-strict	yes	84.3	67.6	86.0	88.5	83.1	76.4	81.0
Qwen3Guard-Stream-4B-loose	yes	83.6	64.3	85.2	88.9	83.3	77.4	80.5
Qwen3Guard-Stream-8B-strict	yes	**85.0**	64.6	85.9	87.5	82.6	77.0	80.4
Qwen3Guard-Stream-8B-loose	yes	84.7	63.1	85.5	88.9	82.4	76.8	80.2
Llama3-StreamGuard-1B **(ours)**	yes	82.4± 0.2	68.0± 0.7	86.2± 0.1	84.6± 0.7	82.9± 0.3	77.2± 0.3	80.2
Llama3-StreamGuard-3B **(ours)**	yes	83.2± 0.3	68.2± 0.2	86.4± 0.0	87.2± 0.2	83.2± 0.2	**77.8**± 0.3	81.0
Llama3-StreamGuard-8B **(ours)**	yes	83.3± 0.3	**69.2**± 0.3	**86.6**± 0.2	**89.9**± 0.5	**84.5**± 0.1	**77.8**± 0.1	**81.9**

Table 7: Output-moderation F1; Avg denotes the macro-average. For prior systems, we report published results. StreamGuard models are evaluated in our simulated streaming protocol, processing responses one token at a time and counting a response as unsafe if it triggers under the debounced rule in § 3.1. We report mean ± std over three seeds for StreamGuard models. The *Streaming* column indicates whether a model supports token-level moderation during generation.

C Dataset-Level Ablation Details

This appendix unpacks the aggregate ablation results in Table 5 into per-benchmark moderation scores. We retain the same Llama3-StreamGuard-8B backbone and the same three practical target-construction factors studied in Section 5: the reduction rule used to aggregate rollout safety scores, the rollout-generator mix across the compared single-model, two-model, and four-model settings, and the supervision-density schedule. These appendix tables should be read as complements to the unified main-text ablation table rather than as separate experiments: Table 5 remains the summary view for streaming intervention and overblocking, while the appendix tables show which standard moderation benchmarks drive the small average differences across variants. In particular, the input-side table is most useful for checking that no single benchmark dominates the macro-average, whereas the output-side table makes clear which datasets absorb the cost of more aggressive streaming operating points.

C.1 Input Moderation

Table 8 reports dataset-level input-moderation F1 for the ablation variants. The central pattern is stability: across reduction rules and rollout mixtures, the macro-average stays in a narrow 87.7–88.2 band, which shows that the small input-side differences in Table 5 are not driven by a single benchmark. No reduction rule dominates the full suite.

Within the reduction block, median reduction improves Aegis to 89.9 and attains the strongest WildGuard score among the reduction variants at 89.6, but gives up performance on OPENAIMODERATION; min reduction is the weakest overall reduction variant, largely because it drops on TOXICCHAT (75.9) and OPENAIMODERATION (73.0). Rollout-mix changes mostly reshuffle where the gains appear: single-model `llama3-8B` rollouts slightly improve Aegis (89.5) and WildGuard (90.0), while the four-model reference remains stronger on TOXICCHAT and OPENAIMODERATION. The supervision-density block produces the largest local swings, especially on policy-sensitive datasets such as TOXICCHAT and OPENAIMODERATION: D32 S8 reaches 79.0 on TOXICCHAT, D16 S8 reaches 76.5 on OPENAIMODERATION, and D1 S8 reaches 90.1 on WildGuard, yet the overall average still

Factor	Variant	ToxiC	OAIMod	Aegis	Aegis2	SSTest	HarmB	WildG	Avg
Reduction	Mean	**77.4** ± 0.5	**75.0** ± 0.4	88.5 ± 0.2	**87.9** ± 0.1	99.5 ± 0.0	99.7 ± 0.1	89.5 ± 0.2	**88.2**
Reduction	Beta Mean	77.0 ± 0.9	**75.0** ± 2.5	88.4 ± 0.3	87.8 ± 0.5	99.5 ± 0.0	99.9 ± 0.1	89.5 ± 0.0	**88.2**
Reduction	Median	77.0 ± 2.0	73.3 ± 1.3	**89.9** ± 0.2	87.7 ± 0.2	99.5 ± 0.0	99.9 ± 0.1	**89.6** ± 0.3	88.1
Reduction	Max	**77.4** ± 0.7	73.6 ± 0.2	89.2 ± 0.5	**87.9** ± 0.5	99.5 ± 0.0	99.9 ± 0.1	89.5 ± 0.3	88.1
Reduction	Min	75.9 ± 1.7	73.0 ± 0.9	88.4 ± 0.7	87.7 ± 0.1	99.5 ± 0.0	99.9 ± 0.1	89.5 ± 0.3	87.7
Rollout Mix	Four-model mixture (16)	**77.4** ± 0.5	**75.0** ± 0.4	88.5 ± 0.2	87.9 ± 0.1	99.5 ± 0.0	99.7 ± 0.1	89.5 ± 0.2	**88.2**
Rollout Mix	Two-model mixture (32)	77.1 ± 0.2	74.3 ± 1.9	88.3 ± 0.0	**88.1** ± 0.1	99.5 ± 0.0	99.9 ± 0.1	89.6 ± 0.1	88.1
Rollout Mix	Single-model (llama3-8B)	76.8 ± 0.3	73.8 ± 1.3	**89.5** ± 0.7	**88.1** ± 0.0	99.5 ± 0.0	99.8 ± 0.0	**90.0** ± 0.1	**88.2**
Rollout Mix	Single-model (qwen2.5 7B)	76.7 ± 1.5	73.8 ± 1.2	88.6 ± 1.1	87.9 ± 0.1	99.5 ± 0.0	99.9 ± 0.1	89.9 ± 0.0	88.0
Supervision Schedule	D64 S{1,2}	77.4 ± 0.5	75.0 ± 0.4	88.5 ± 0.2	87.9 ± 0.1	99.5 ± 0.0	99.7 ± 0.1	89.5 ± 0.2	88.2
Supervision Schedule	D64 S4	78.0	75.1	88.4	87.8	99.5	99.4	89.9	88.3
Supervision Schedule	D64 S8	78.1	75.9	88.7	87.8	99.5	99.8	89.5	**88.5**
Supervision Schedule	D32 S4	77.6	75.1	88.7	**88.0**	99.5	**100.0**	89.4	88.3
Supervision Schedule	D32 S8	**79.0**	75.7	88.4	87.9	99.5	99.4	89.7	**88.5**
Supervision Schedule	D16 S4	76.7	76.1	88.7	87.7	99.5	100.0	89.7	88.3
Supervision Schedule	D16 S8	79.3	**76.5**	88.1	87.9	99.5	98.3	89.6	88.4
Supervision Schedule	D1 S4	78.4	74.1	88.5	87.8	99.5	99.4	89.2	88.1
Supervision Schedule	D1 S8	76.8	75.3	**89.2**	87.8	99.5	**100.0**	**90.1**	88.4

Table 8: Dataset-level input-moderation breakdown for the ablations in Table 5. This appendix table decomposes the aggregate input-moderation result from the main text into per-benchmark F1 scores for the same Llama3-StreamGuard-8B variants, varying the reduction rule used to aggregate rollout oracle scores, the rollout-generator mix across the compared single-model, two-model, and four-model settings, and the supervision-density schedule. Consistent with the paper's main ablation story, input-side moderation remains tightly clustered across these practical target-construction choices, indicating that forecasting-based supervision is robust rather than dependent on a brittle implementation detail. Avg denotes the macro-average across benchmarks; supervision-schedule notation follows Table 5; reported values are mean ± standard deviation over three seeds where shown.

remains between 88.1 and 88.5. These details reinforce the main-text interpretation that forecasting-based supervision is robust on standard input moderation, with most practical variants shifting where the model is slightly more or less strict rather than changing the overall quality level.

C.2 Output Moderation

Table 9 reports dataset-level output-moderation F1 under the same streaming evaluation protocol used throughout the paper. This decomposition makes the main trade-off more concrete.

Within the reduction block, the mean-reduction reference gives the best macro-average (81.9) because it is consistently strong across BEAVERTAILS (86.6), XSTEST (89.9), and WildGuard (77.8), even though individual alternatives win isolated columns. Max reduction is the clearest example of the aggressive-intervention trade-off: it slightly improves SAFERLHF to 69.8 and gives the strongest streaming numbers in Table 5, but gives up substantial performance on XSTEST (85.9) and WildGuard (71.2), which explains its weaker 79.0 output-side average. Min reduction shows the opposite operating point, doing best on HARMBENCH (84.7) and AEGIS2.0 (85.0) while dropping sharply on SAFERLHF (63.3). The rollout-mixture block tells a similar story. The four-model reference is strongest or tied on SAFERLHF, BEAVERTAILS, XSTEST, and WildGuard, whereas single-model `llama3-8B` rollouts slightly improve HARMBENCH (83.6) but lose on AEGIS2.0 and WildGuard; the single-model `qwen2.5-7B` rollout pool underperforms most clearly on SAFERLHF (65.7) and WildGuard (75.2).

Supervision density again changes the operating point more than the overall quality: D64 S4 improves XSTEST to 90.6 and WildGuard to 78.9, D32 S8 matches the best average at 82.0 with the strongest BEAVERTAILS score (86.8), and D1 S4 preserves strong SAFERLHF performance (69.6) but drops notably on XSTEST (88.1) and WildGuard (75.4). Read together with the streaming and FPR columns in Table 5, these dataset-level shifts make clear that more aggressive variants do not degrade uniformly; they lose most on benchmarks that reward benign calibration or broader output-side robustness.

Factor	Variant	HarmB	SRLHF	BeaverTails	XSTESTR	Aegis2	WildG	Avg
Reduction	Mean	83.3 ±0.3	69.2 ±0.3	**86.6** ±0.2	**89.9** ±0.5	84.5 ±0.1	**77.8** ±0.1	**81.9**
Reduction	Beta Mean	83.4 ±0.7	67.6 ±0.5	86.5 ±0.2	89.2 ±0.7	84.7 ±0.8	77.3 ±1.1	81.4
Reduction	Median	83.5 ±0.6	67.3 ±1.0	86.1 ±0.0	88.6 ±0.2	84.3 ±0.9	77.2 ±0.5	81.2
Reduction	Max	79.8 ±0.2	**69.8** ±0.8	85.9 ±0.2	85.9 ±0.2	81.1 ±0.5	71.2 ±1.3	79.0
Reduction	Min	**84.7** ±0.1	63.3 ±0.7	85.6 ±0.2	88.4 ±0.9	**85.0** ±0.6	78.6 ±0.8	80.9
Rollout Mix	Four-model mixture (16)	83.3 ±0.3	**69.2** ±0.3	**86.6** ±0.2	**89.9** ±0.5	84.5 ±0.1	**77.8** ±0.1	**81.9**
Rollout Mix	Two-model mixture (32)	83.3 ±0.1	67.4 ±0.2	86.0 ±0.0	89.4 ±0.0	84.0 ±0.5	76.5 ±1.0	81.1
Rollout Mix	Single-model (llama3-8B)	**83.6** ±0.4	68.9 ±0.5	86.2 ±0.1	88.5 ±2.3	82.6 ±0.3	75.5 ±1.2	80.9
Rollout Mix	Single-model (qwen2.5 7B)	82.9 ±0.7	65.7 ±0.5	85.6 ±0.2	88.6 ±2.0	84.3 ±0.2	75.2 ±0.2	80.4
Supervision Schedule	D64 S{1,2}	83.3 ±0.3	69.2 ±0.3	86.6 ±0.2	89.9 ±0.5	84.5 ±0.1	77.8 ±0.1	81.9
Supervision Schedule	D64 S4	**84.1**	69.3	86.5	**90.6**	82.7	**78.9**	**82.0**
Supervision Schedule	D64 S8	**84.1**	68.9	86.5	89.7	84.2	78.3	81.9
Supervision Schedule	D32 S4	82.7	69.2	86.4	89.1	84.6	77.4	81.6
Supervision Schedule	D32 S8	83.9	69.3	**86.8**	90.2	84.2	77.8	**82.0**
Supervision Schedule	D16 S4	84.3	69.4	86.7	89.5	84.2	77.4	81.9
Supervision Schedule	D16 S8	83.8	69.7	86.6	89.5	84.8	77.6	**82.0**
Supervision Schedule	D1 S4	82.7	**69.6**	86.3	88.1	84.4	75.4	81.1
Supervision Schedule	D1 S8	83.1	69.5	86.6	89.5	**85.8**	77.1	81.9

Table 9: Dataset-level output-moderation breakdown for the ablations in Table 5. This appendix table decomposes the aggregate output-moderation result from the main text into per-benchmark F1 scores under the same streaming evaluation protocol used throughout the paper, where a response is marked unsafe if the guardrail triggers at any point during generation. Read together with the main ablation table's streaming-timing and overblocking columns, these results support the central ablation story: StreamGuard remains competitive across a broad range of target-construction variants, while the meaningful differences come from the operating-point trade-off between more aggressive early intervention and better benign calibration. Avg denotes the macro-average across benchmarks; supervision-schedule notation follows Table 5; reported values are mean ± standard deviation over three seeds where shown.

D Cross-Tokenizer Transfer Details

This appendix provides the dataset-level breakdown underlying the aggregated transfer results in Table 4. We retain the same native Llama3-StreamGuard-1B reference and the same transferred Gemma and Qwen models discussed in Section 4. The two appendix tables separate a pattern that is compressed in the main transfer table: input-side transfer is somewhat uneven across benchmark families, whereas output-side transfer is substantially more consistent and accounts for most of the gain on streaming-related metrics.

D.1 Input Moderation

Table 10 reports per-benchmark input-moderation F1 for the native and transferred models. Input-side transfer is possible, but it is less uniform than output-side transfer.

The native Llama3-StreamGuard-1B reference remains strongest on the macro-average (86.4), driven by clear leads on Aegis (89.0), AEGIS2.0 (87.7), and WildGuard (88.6). The transferred Qwen3-StreamGuard-1.7B model comes closest at 85.8 and actually exceeds the native model on SIMPLESAFETYTESTS (98.8 vs. 98.6) and HARMBENCH (99.8 vs. 94.8), but it trails on Aegis, AEGIS2.0, and WildGuard, which keeps its overall average below the native reference. Gemma3-StreamGuard-1B reaches 85.1, with near-reference performance on TOXICCHAT, OPENAIMODERATION, AEGIS2.0, and WildGuard, but lower scores on Aegis and especially HARMBENCH.

The smaller transferred models illustrate the capacity limitation more clearly: Gemma3-StreamGuard-0.3B falls to 79.9 on the macro-average, with especially weak TOXICCHAT/OPENAIMODERATION results, which indicates that transfer alone does not compensate for limited model capacity. Overall, the input-side table supports the main-text claim that transfer is possible but less uniform on standard input moderation than on output-side or streaming behavior.

model	Cross-Tokenizer	ToxiC	OAIMod	Aegis	Aegis2	SSTest	HarmB	WildG	Avg
Qwen3Guard-Stream-0.6B-strict	No	72.0	68.3	85.2	84.9	98.0	97.2	87.1	84.7
Qwen3Guard-Stream-0.6B-loose	No	**75.5**	**76.0**	77.7	81.7	96.9	96.8	86.0	84.4
Llama3-StreamGuard-1B **(ours)**	No	74.9± 0.3	71.5± 0.2	**89.0**± 0.3	**87.7**± 0.2	98.6± 0.2	94.8± 2.9	**88.6**± 0.2	**86.4**
Gemma3-StreamGuard-0.3B **(ours)**	Yes	60.1 ±3.0	63.8 ±0.6	83.9 ±0.5	82.9 ±0.5	96.0 ±0.6	89.1 ±4.5	83.7 ±0.7	79.9
Gemma3-StreamGuard-1B **(ours)**	Yes	74.0 ±0.4	71.2 ±0.4	87.4 ±1.3	86.6 ±0.2	98.5 ±0.0	89.5 ±2.8	88.4 ±0.2	85.1
Qwen3-StreamGuard-0.6B **(ours)**	Yes	66.9 ±0.8	67.7 ±0.8	86.0 ±0.3	85.1 ±0.3	96.9 ±0.5	88.1 ±0.9	85.8 ±0.4	82.3
Qwen3-StreamGuard-1.7B **(ours)**	Yes	72.1 ±1.9	71.1 ±0.6	85.3 ±0.9	85.7 ±0.4	**98.8** ±0.3	**99.8** ±0.2	87.9 ±0.3	85.8

Table 10: Dataset-level input-moderation results for the cross-tokenizer transfer setting, complementing Table 4. "Cross-Tokenizer" indicates whether prefix-level future-risk supervision is transferred from a source model with a different tokenizer than the target model. Despite the tokenizer mismatch, transferred StreamGuard models remain competitive on standard input-moderation benchmarks. Avg denotes the macro-average across datasets; reported values for StreamGuard models are mean ± standard deviation over three seeds where shown.

model	Cross-Tokenizer	HarmB	SRLHF	BeaverTails	XSTestR	Aegis2	WildG	Avg
Qwen3Guard-Stream-0.6B-strict	No	**83.1**	62.8	84.5	84.8	81.4	76.3	78.8
Qwen3Guard-Stream-0.6B-loose	No	80.6	61.7	84.0	83.3	81.4	75.8	77.8
Llama3-StreamGuard-1B **(ours)**	No	82.4 ± 0.2	68.0 ± 0.7	86.2 ± 0.1	84.6 ± 0.7	82.9 ± 0.3	**77.2** ± 0.3	80.2
Gemma3-StreamGuard-0.3B (**ours**)	Yes	77.2 ±2.8	65.6 ±0.4	85.2 ±0.4	76.1 ±2.4	80.7 ±0.6	67.1 ±3.4	75.3
Gemma3-StreamGuard-1B (**ours**)	Yes	82.7 ±0.8	69.7 ±0.4	**87.2** ±0.2	**87.2** ±0.0	**84.7** ±0.3	76.3 ±1.0	**81.3**
Qwen3-StreamGuard-0.6B (**ours**)	Yes	80.9 ±0.9	**68.8** ±0.4	86.3 ±0.4	82.8 ±0.8	83.2 ±0.7	74.4 ±0.8	79.4
Qwen3-StreamGuard-1.7B (**ours**)	Yes	82.8 ±1.1	**68.8** ±1.3	86.8 ±0.2	83.0 ±1.7	83.1 ±0.4	76.0 ±1.5	80.1

Table 11: Dataset-level output-moderation results for the cross-tokenizer transfer setting, complementing Table 4. Models are evaluated under the same streaming output-moderation protocol used in the main paper, where a response is marked unsafe if it triggers at any point during generation. Cross-tokenizer transfer remains effective across target families, with transferred StreamGuard models preserving strong output-moderation performance. Avg denotes the macro-average across datasets; reported values for StreamGuard models are mean ± standard deviation over three seeds where shown.

D.2 Output Moderation

Table 11 reports per-benchmark output-moderation F1 under the same streaming evaluation protocol used throughout the paper. The output-side pattern is more consistent.

Gemma3-StreamGuard-1B achieves the best transferred macro-average (81.3), driven by the best BEAVERTAILS (87.2), XSTEST (87.2), and AEGIS2.0 (84.7) scores together with a strong SAFERLHF result (69.7). In the main transfer table, it also gives the best transferred streaming F1 (98.2) and miss rate (3.5%), although its OnTime rate (92.3%) is slightly below the native Llama3-StreamGuard-1B reference (92.9%). The native Llama3-StreamGuard-1B reference remains best on WildGuard (77.2), but is otherwise matched or exceeded by the stronger transferred backbones. The transferred Qwen models are also competitive: both Qwen variants reach 68.8 on SAFERLHF and remain close to the native model on HARMBENCH and BEAVERTAILS, though they are weaker on XSTEST and WildGuard than Gemma3-StreamGuard-1B.

The small Gemma3-StreamGuard-0.3B model again shows that transfer still depends on model capacity: it retains reasonable BEAVERTAILS performance (85.2) but drops sharply on XSTEST (76.1) and WildGuard (67.1), pulling its average down to 75.3. These finer-grained results support the main practical takeaway: future-risk supervision transfers well across tokenizer families for output moderation, but the strength of the transfer still depends on target-model capacity and family-specific calibration.

E Latency Measurements

This appendix reports latency measurements for streaming moderation. Our main setting is *pipelined* moderation, in which the guardrail runs concurrently with decoding. In this regime, the key systems question is whether guardrail decisions can keep pace with the generator's token cadence, and therefore whether risky output can be blocked before additional tokens